



Strathmore
UNIVERSITY

School of Computing and Engineering Sciences
Bachelor of Science in Computer Networks and Cybersecurity
Semester 2 (Aug 2025 – Dec 2025)

Course Outline

Unit Code: CNS 2203

Unit Name: Data Network Design and Management II

Lecturer: Tiberius T. Nkinyili.

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First Floor

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Module Leader:

Purpose of the Unit

To provide students with knowledge and skills in the design and implementation of data networks.

Overall Learning Outcomes

At the end of this course, the student will be able to:

- i. Apply computer networks principles.
- ii. Apply routing and translation
- iii. Demonstrate how networks are integrated into business practices.
- iv. Design networks using traffic modelling.
- v. Apply gained skills for network design in a heterogeneous environment.

Contact Hours

60 Hours

Pre-Requisites

CNS 2104: Data Network Design I			
Delivery Outline			
Week	Module	Sub-topic	Description
1	Introduction Class Course Outline Review and recap	- Semester Expectations - Review Switching and Basic Switch Security	Comprehensive Lab and Discussion Q & A
2	Review Etherchannel and Wireless LANs	- Discuss Etherchannel concepts and configurations - Discuss WLAN architectures - Configure WLANs	Lectures Packet Tracer configuration Q & A
3	Introduction to Routing Concepts and Configuration of Static Routing	- Discuss Routing Concepts - Analyse Routing Tables - Review Routing processes, static and dynamic - Review Types of Static routes	Lectures Packet Tracer configuration Q & A
4	Dynamic Routing and RIP	- Discuss route summarization - Configure RIP - Configure RIPng (for IPv6)	Discussions Static route configuration RIP and RIPng configuration Packet Tracer labs
5-6	OSPFv2	- Introduction to OSPFv2 concepts, features, and characteristics - OSPF Operations. - OSPF V2 configuration- single area OSPF - Modify Single-Area	Discussions on OSPF versions Q & A Configure OSPFv2

		OSPFv2,	
7	OSPFv3	• OSPF router ID • Point to point OSPF networks • Multi-Access OSPF • Route propagation and verification	OSPFv3 Lab and PT Q & A Discussion Assignment- BGP/EIGRP
8-9	ACLs	• Introduction to ACLs: Purpose and types of IPv4 ACLs • Wildcard Masks in ACLs • ACL Creation • Configure and modify standard IPv4 ACLs. • Apply ACLs to interfaces • Secure VTY ports with a Standard IPv4 ACLs.	Cases and labs PT activities Q & A Discussion
10-11	NAT & PAT	• Introduction to NAT: Characteristics, Types • Static NAT • Dynamic NAT • Port Address Translation (PAT) • NAT 64	Q & A Discussion NAT config on PT Dynamic NAT and PAT config lab
12	Network Management and Troubleshooting	• QoS • SNMP • Logging and Configuration Management	Q & A Discussion QoS labs Configure SNMP and Logging
13	Tutorials (Recap and Revision)		
Mode of Delivery			

Lectures, Tutorials and Labs

Core Reading Materials

1. CCNAv7 Enterprise Networking, Security, and Automation
2. Computer Networking: A Top-Down Approach (7th Edition), James Kurose and Keith Ross, Pearson, 7th ed. (2016), ISBN-10: 0133594149 (ISBN-13: 978-0133594140)

Recommended Reference Materials

1. *Computer Networking: A Top Down Approach*, James F., Keith W., Pearson; 7th ed (2017) ISBN 9780133594140
2. Data and Computer Communications, W. Stallings, Prentice Hall, 10th ed. (2013), ISBN-10: 0133506487 ISBN-13: 978-0133506488
3. Wireless Communications & Networking, Vijay Garg, Morgan Kaufmann, 1st ed. (2007) ISBN-10: 0123735807 ISBN-13: 978-0123735805
4. Network Analysis, Architecture and Design, McCabe, J. D., Morgan-Kaufmann (2003), 1558608877, 978-1558608877
5. Wireless Communications & Networks, William Stallings, Prentice Hall, 2nd ed. (2004) ISBN-10: 0131918354 ISBN-13: 978-0131918351

E-resources

Cisco Networking Academy-www.netacad.com

Journals

IEEE Xplore -<https://ieeexplore.ieee.org.ezproxy.library.strathmore.edu>

Unit Assessment Type	Weighting (%)
Final examination	50
CAT 1	10
CAT 2	10
Assignments	10
Labs	20
Total	100

Lectures/Classes

1. Punctuality is **Important**
2. Active participation in class discussions is encouraged

Assignments and/or Coursework

1. **Plagiarism** is a serious offence. If detected in any form the following will apply:
 - a. In partial cases (such as not citing whole word-for-word quotes), half the total possible marks of the assignment shall be duly struck off
 - b. In serious cases (such as whole duplication of an assignment), a zero policy will apply, i.e., all offending assignments shall be awarded a mark of zero.

Note: The level of seriousness referred to above is at the discretion of the lecturer. Appeals are certainly possible through the relevant channels.
2. Notwithstanding the above, **collaboration** in coursework is encouraged as this promotes team spirit and group synergy (as long as originality is preserved).
3. If you consult other references for an assignment, you must cite them using the 7th edition of the APA style.

Communication Channels

1. e-mail (**Maintain formality**)
2. Unit leader-
3. Google Classroom platform-